

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

COURSE OUTCOME COVERED

C0-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool: Application-Based Questions

Weightage: 40%

QUESTIONS: Total Marks: 30

1: Online Symposium Portal

A college is developing an online symposium registration portal. The following HTML structure and HTTP interaction is observed:

```
<form action="/register" method="POST">
  <input type="text" name="name">
  <input type="email" name="email">
  <button>Submit</button>
</form>
```

When a student submits the form, the browser sends an HTTP request to the server, and the server responds accordingly.

Questions:

1. Identify appropriate **HTML5 semantic elements** missing in the structure.
2. Modify the structure to make it **standards-compliant**.
3. Interpret the **HTTP request generated** during form submission.
4. Analyze the **HTTP response cycle** from server to browser.

5. Suggest improvements for **usability and accessibility**.

2: Cross-Browser Compatibility Issue

A company website shows inconsistent layout across browsers. The following HTML snippet is used:

```
<html>
<head>
<title>Company</title>
</head>
<body>
<div>
<h1>Welcome
<p>About us
</div>
</body>
</html>
```

Questions:

1. Identify **improper nesting and missing closing tags**.
2. Analyze how different browsers may interpret this structure.
3. Identify **missing semantic elements** (e.g., header, section).
4. Provide a **corrected W3C-compliant HTML structure**.
5. Suggest techniques to ensure **cross-browser compatibility**.

3: Form Submission Failure

A web application fails to send form data to the server. The following configuration is observed:

```
<form action="/submit" method="GET">
  <input type="text" name="username">
  <input type="password" name="password">
  <button type="button">Login</button>
</form>
```

Questions:

1. Identify issues in the **form configuration**.
2. Analyze why the **HTTP request is not triggered**.
3. Interpret the role of **GET method in this scenario**.
4. Propose a **corrected implementation**.
5. Suggest improvements for **secure data transmission**.

Code of Conduct:

I T. Rupendra/192311325 (Name I Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary

action.

Signature of the student: T. Rupendra

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level4 – Exemplary	Level3 – Proficient	Level2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Answers

I)Online Symposium Portal

1. Identify missing HTMLS semantic elements

- `<html>`
- `<head>`
- `<title>`
- `<main>`
- `<section>`
- `<label>` for form fields

2. Standards-compliant HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Symposium Registration</title>
</head>
<body>
<main>
<section>
<h2>Registration Form</h2>

<form action="/register" method="POST">
  <label for="name">Name</label>
  <input type="text" id="name" name="name" required>

  <label for="email">Email</label>
  <input type="email" id="email" name="email" required>

  <button type="submit">Submit</button>
</form>

</section>
</main>
</body>
</html>
```

3. HTTP Request

- Method: POST
- URL: `/register`
- Data sent:

name=John

email=john@example.com

4. HTTP Response Cycle

1. Browser sends POST request.
2. Server processes the data.
3. Server returns a response (e.g., 200 OK or 302 Redirect).
4. Browser displays confirmation page.

5. Usability & Accessibility Improvements

- Use `<label>` elements.
- Add `required` validation.
- Use semantic tags (`<main>`, `<section>`).
- Provide clear button text.
- Ensure keyboard accessibility.

2. Cross-Browser Compatibility Issue

1. Improper Nesting & Missing Closing Tags

- `<h1>` not closed.
- `<p>` not closed.
- Missing `<!DOCTYPE html>`.

2. Browser Interpretation

- Browsers may automatically close missing tags.
- Different browsers may display different layouts.

3. Missing Semantic Elements

- `<header>`
- `<main>`
- `<section>`

4. Correct W3C-Compliant HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Company</title>
</head>

<body>

<header>
<h1>Welcome</h1>
</header>

<main>
<section>
<p>About us</p>
</section>
</main>

</body>
</html>
```

5. Cross-Browser Compatibility Techniques

- Use valid HTML5.
- Test on multiple browsers.
- Use CSS Reset/Normalize.
- Validate HTML and CSS.
- Use responsive design.

3. Form Submission Failure

1. Issues in Form Configuration

- Uses **GET** instead of **POST**.
- Button type is "**button**" instead of "**submit**".

2. Why HTTP Request is Not Triggered

- `type="button"` does not submit the form.
- Only `type="submit"` sends the request.

3. Role of GET Method

- Sends data in the URL.
- Suitable for search forms.
- Not secure for passwords.

4. Correct Implementation

```
<form action="/submit" method="POST">
```

```
<label>Username</label>
```

```
<input type="text" name="username" required>
```

```
<label>Password</label>
```

```
<input type="password" name="password" required>
```

```
<button type="submit">Login</button>
```

```
</form>
```

5. Secure Data Transmission Improvements

- Use **POST** method.
- Use **HTTPS**.
- Validate input on client and server.
- Hash passwords before storing.
- Enable authentication and session management.